

Knowledge Alignment Benchmarking: Evaluating Causal Clinical Reasoning in Multimodal LLMs on Brain MRI

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Abstract

Multimodal large language models (MLLMs) are increasingly evaluated on medical imaging tasks, but almost every existing benchmark scores only final-answer accuracy. Accuracy alone cannot distinguish a model that reasoned correctly from one that pattern-matched the question text or produced reasoning that contradicts its own answer – a distinction that matters in clinical settings, where an unfaithful but accidentally correct answer will not generalize to a novel case. We introduce **Knowledge Alignment Benchmarking (KAB)**, a framework for evaluating multimodal clinical reasoning as a causal chain of five phases mirroring how a radiologist works through a brain MRI case: anatomical assessment, lesion localization, diagnostic synthesis, prognostic judgment, and treatment planning. KAB scores models along three axes beyond accuracy: *faithfulness*, measured by grounding a model’s stated reasoning against two authoritative medical ontologies (RadLex and the NCI Thesaurus) and by probing whether a model’s reasoning text alone predicts its own answer; *feedback-adaptation*, measured by injecting a model’s missed clinical concepts and re-querying to see whether it self-corrects; and *soft gating*, which blocks a phase from being scored as valid reasoning if the phases it causally depends on were not themselves answered correctly. In building this framework on the existing OmniBrainBench dataset, we found that its phase groupings do not reflect genuine same-patient continuity – only 2 of 6,823 questions in the dataset form a real same-patient multi-phase pair – which undermines any claim about reasoning degrading across a single patient’s causal chain. We describe the construction of a genuine same-patient longitudinal chain dataset built on LUMIERE, a real 91-patient glioblastoma cohort, using a hybrid pipeline that drafts multiple-choice items from patient facts with an LLM and validates every item with a domain-expert clinical review. This pa-

per presents the KAB framework, the construct-validity problem it surfaces in existing benchmarks, and the dataset construction method used to address it; full evaluation results across the model suite are reported in a companion release.

1 Introduction

Multimodal large language models (MLLMs) are being applied to medical imaging tasks – reading a brain MRI, describing a lesion, suggesting a diagnosis – at a pace that has outrun the tools available to evaluate them safely. The dominant evaluation paradigm treats a medical visual question answering (VQA) task exactly like any other multiple-choice benchmark: a model is scored on whether it selects the correct option, and nothing else is measured. This is a critical gap for clinical applications, because final-answer accuracy cannot distinguish between three behaviors that look identical at the level of a selected letter but are clinically distinct:

1. The model correctly perceived the imaging findings and reasoned validly to a diagnosis or plan.
2. The model pattern-matched superficial cues in the question text or answer options, arriving at the right answer without engaging with the image at all.
3. The model guessed, or produced a chain of reasoning that is internally inconsistent with the answer it ultimately gave.

Behaviors (2) and (3) are well documented outside the medical domain: crowd-sourced natural language inference datasets contain annotation artifacts that let models reach high accuracy from the hypothesis sentence alone (?), and models trained on such data adopt syntactic heuristics that collapse under adversarial evaluation (?). Chain-of-thought reasoning inherits the same failure mode:

a model’s stated reasoning frequently does not reflect the computation that actually produced its answer (?); this occurs on naturally worded prompts without any adversarial construction (?), and intermediate reasoning structures behave as influential context rather than as stable causal mediators of the final decision (?). In the medical multimodal setting specifically, cascading errors from early-stage reasoning failures have been identified as a leading cause of incorrect predictions (?), and models have been shown to rely on textual or visual shortcuts rather than the pathological regions a question actually asks about (?). In a clinical setting, an accurate-but-unfaithful answer is not merely a benchmark artifact – it is a model that cannot be trusted on the next case that differs even slightly from the ones it was evaluated on.

We argue that a medical MLLM benchmark should evaluate reasoning the way a clinician’s diagnostic process is actually structured: as a sequential, error-compounding chain, where later judgments are only valid if earlier ones were correct (?). We introduce **Knowledge Alignment Benchmarking (KAB)**, a framework that decomposes a brain MRI case into five clinical phases – Anatomical and Imaging Assessment (AIA), Lesion Identification and Localization (LIL), Diagnostic Synthesis and Causal Reasoning (DSCR), Prognostic Judgment and Risk Forecasting (PJRF), and Therapeutic Cycle Management (TCM) – and scores a model on three dimensions beyond accuracy: whether its reasoning is grounded in real clinical vocabulary and is self-consistent (**faithfulness**), whether it can incorporate a targeted clinical correction when it fails (**feedback-adaptation**), and whether its chain of answers is causally coherent rather than a set of independently-guessed phases (**soft gating**). Figure ?? summarizes the five-phase chain and how each of the three metrics attaches to it.

In the course of building KAB on top of the existing OmniBrainBench dataset (?), we discovered a construct-validity problem that we believe generalizes beyond our own project: OmniBrainBench’s phase-to-case grouping does not track same-patient identity. Of 6,823 questions spanning 15 source sub-corpora, only 2 form a genuine same-patient pair across two different clinical phases, and zero span three or more phases. A "chain" evaluated under the current grouping is stitched together from different real patients’ questions within the same origin sub-corpus, so a model being "gated out" of a later phase after failing an earlier one

is an artifact of this aggregation, not evidence of causal reasoning failure for one patient. We report this problem in detail (Section ??) and describe our resolution: a genuine same-patient longitudinal chain dataset built on LUMIERE (?), a 91-patient real-world glioblastoma cohort with longitudinal imaging, expert-assigned RANO treatment-response ratings, survival data, and Stupp-protocol treatment records, using a hybrid LLM-draft-plus-expert-review construction pipeline (Section ??).

This paper makes four contributions. First, we identify and quantify a construct-validity problem in an existing brain-MRI MLLM benchmark that is likely to recur in any dataset assembled from single-task source corpora. Second, we present a reusable hybrid construction method – automatic fact extraction and LLM item drafting followed by targeted domain-expert review – for turning a raw longitudinal clinical imaging dataset into a validated multi-phase reasoning benchmark. Third, we present the KAB framework itself: ontology-grounded faithfulness scoring against RadLex and the NCI Thesaurus, a feedback-adaptation loop that distinguishes a knowledge gap from an activation failure, and soft gating that operationalizes the causal dependency between clinical phases. Fourth, once chains are genuine, KAB’s degradation and adaptation measurements become answers to a well-posed question – whether a model’s reasoning at a later phase actually follows from what it established earlier for the same patient – rather than an artifact of how a dataset happens to be grouped. Full evaluation results across our model suite are reported in a companion release, as data collection is ongoing at the time of writing (Section ??).

2 Related Work

Shortcut learning and reasoning artifacts. The observation that a model can reach a correct label without engaging with the intended reasoning task predates large language models. ? showed that natural language inference hypotheses alone – without the premise – carry enough annotation artifacts to substantially outperform chance. ? showed that models trained on such data adopt syntactic heuristics (lexical overlap, subsequence, constituent) that fail systematically on their HANS diagnostic set. KAB’s local faithfulness probe and correct-but-unfaithful flag are a direct descendant of this line of work, applied to a setting where the "hypothesis" is a model’s own free-text reasoning rather than a

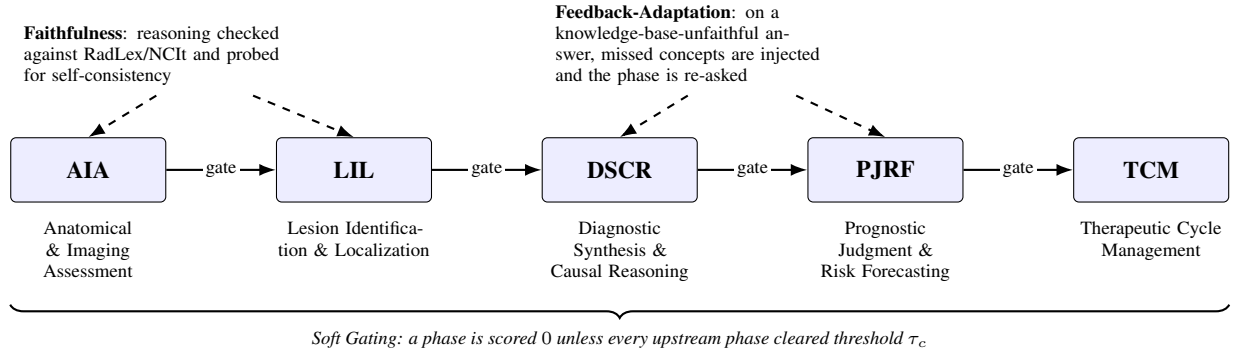


Figure 1: The KAB framework: a brain-MRI case is evaluated as a five-phase causal chain (bottom), with three metrics attached to it (top and below) – faithfulness (grounding + self-consistency), feedback-adaptation (a targeted re-query on failure), and soft gating (a phase counts only if its upstream dependencies were answered correctly).

dataset artifact.

Chain-of-thought faithfulness. ? introduced systematic tests for whether a model’s chain-of-thought causally affects its final answer, finding that stated reasoning frequently does not reflect what actually drove the answer. ? extended this to naturally-worded, non-adversarial prompts, reporting unfaithfulness rates up to 13% in production models. ? evaluated schema-guided pipelines under causal intervention on intermediate structures and found models fail to update their final decision after the intervention in up to 60% of cases, concluding that intermediate reasoning functions as influential context rather than a stable causal mediator. KAB’s local faithfulness probe is methodologically closest to this line: rather than intervening on the reasoning text, we withhold everything except the reasoning text and ask whether it alone reproduces the original answer.

Cascading failure in medical multimodal reasoning. Two recent works are close enough to KAB’s gating mechanism to warrant explicit comparison. ? propose Medical Reasoning-aware Policy Optimization (MRPO), a reinforcement-learning method that assigns exponentially larger penalties to earlier invalid reasoning steps during training, directly targeting the same cascading-failure phenomenon that motivates KAB’s soft gating. ? introduce perturbation tests that detect textual and visual shortcut learning in medical VQA models. Both works target the same underlying phenomenon – early-stage reasoning failure propagating into later errors – but neither addresses *dataset construct validity*: whether the multi-step cases used for training or evaluation actually represent one patient’s causal trajectory in the first place. MRPO is a

training-time intervention that presupposes its step-labeled data is a genuine causal chain; our contribution is upstream of that assumption. KAB additionally couples cascading-failure detection to ontology-grounded faithfulness scoring and a feedback-adaptation loop, neither of which is part of either prior work.

Clinical cognition as motivation for phase structure. The five-phase decomposition in KAB is not an arbitrary engineering choice. ? describes clinical diagnostic reasoning under a dual-process model in which sequential, error-compounding cognitive steps – rather than a single holistic judgment – characterize real diagnostic error. This gives ²⁰⁰ clinical-science grounding for treating AIA $\rightarrow$ LIL $\rightarrow$ DSCR $\rightarrow$ PJRF $\rightarrow$ TCM as a genuine dependency chain rather than a convenient dataset partition.

Brain MRI benchmarks and same-patient continuity. ²⁰⁴OmniBrainBench (?) is, to our knowledge, the largest closed-ended brain-MRI MLLM benchmark organized around a multi-phase clinical workflow, and is the dataset on which we build the KAB framework. ²⁰⁵NeuroQA (?) is substantially larger (56,953 QA pairs, 12,977 subjects) and includes a "Longitudinal" question category, but this category compares two timepoints rather than a full causal chain, and its authors explicitly state the benchmark is not intended to support claims about treatment planning. UCSF-PDGM-VQA (?) provides 2,387 QA pairs over 473 glioma studies but at a single timepoint per patient, with no prognostic or treatment layer. LUMIERE (?) and MU-Glioma-Post (?) both provide genuine longitudinal, real-patient continuity with treatment and outcome data, but neither ships a question-answering layer. We are

not aware of a published dataset that combines genuine same-patient continuity across imaging, diagnosis, prognosis, and treatment with a ready-made multiple-choice question layer; Section ?? describes our construction of one on top of LUMIERE.

3 The KAB Framework

KAB evaluates a model on a per-case basis, where a case is one patient’s progression through some subset of the five clinical phases. For each question in a case, a model receives the relevant image(s), the question, the answer options, and a structured summary of its own answers from upstream phases in the same case (the *chain context*). It must return a structured response containing an answer, a description of what it observed in the image (visual_grounding), and free-text reasoning connecting the observation to the answer. On top of this base interaction, KAB computes three families of metrics.

3.1 Faithfulness

Local faithfulness. After a model answers, we issue a second call – the faithfulness probe – containing *only* the model’s own reasoning text (no image, no question, no answer options) and ask which answer that reasoning supports. If the probe’s predicted answer matches the model’s original answer, the response is *locally faithful*: the stated reasoning is sufficient, on its own, to recover the conclusion. A mismatch means the model’s justification is disconnected from the answer it actually gave. We separately flag *correct-but-unfaithful* responses – a right answer paired with reasoning that does not support it – as the most clinically dangerous category, since such a response offers no evidence it will generalize to a case that differs even slightly from the one it was scored on.

KB alignment score. This is KAB’s central grounding metric. Every diagnostic phase has an authoritative source of real clinical vocabulary: RadLex (?) for radiology anatomy, imaging modalities, and lesion descriptors (routed to the AIA and LIL phases), and the NCI Thesaurus (NCIt) for tumor classification, WHO grading, molecular markers, staging, and treatment (routed to DSCR, PJRF, and TCM). NCIt was chosen over UMLS because OmniBrainBench and LUMIERE are both predominantly glioma/glioblastoma datasets, for which NCIt has the deepest freely-available cov-

erage, and because it requires no account or API key, removing a reproducibility barrier. We concatenate a model’s visual_grounding and reasoning fields, tokenize the text (preserving hyphenated compounds such as *ring-enhancing* as single tokens), and run a greedy longest-match scan against the union of both knowledge bases, trying 4-grams down to 1-grams with no reuse of consumed token positions. This yields extracted_concepts – every KB term the model used regardless of phase. We then take matched_concepts as the subset of extracted_concepts present in the phase-appropriate KB, and compute

$$\text{KBAlign} = \frac{|\text{matched_concepts}|}{|\text{extracted_concepts}|}. \quad (1)$$

A low score indicates the model is not using vocabulary appropriate to the current diagnostic phase – for example, still describing anatomy at the diagnosis phase rather than committing to tumor-classification language.

Concept precision score. KB alignment measures whether a model used *real* clinical terms; concept precision measures whether it used the *right* ones for the specific case. Of the matched phase-KB concepts, we compute the fraction that also appear in the correct answer text:

$$\text{ConceptPrecision} = \frac{|\text{on-target matches}|}{|\text{matched_concepts}|}. \quad (2)$$

A model can score high on KB alignment while scoring low on concept precision – using plausible-sounding real clinical language that is nonetheless wrong for the case at hand.

Chain faithfulness. For each phase after AIA, we check what fraction of the key terms from the correct answers of *upstream* phases are referenced in the current phase’s reasoning text. A model that identified a lesion in the right temporal lobe during LIL but never references that location during DSCR is answering each phase in isolation rather than reasoning through a coherent patient-level chain.

3.2 Feedback-Based Adaptation

Faithfulness metrics establish whether a model reasoned correctly; the adaptation loop asks a different question – whether a model that failed can be corrected. The loop triggers when a response is KB-unfaithful (KBAlign below a threshold τ_B , default 0.3) and has a non-empty set

of missed concepts, defined as phase-KB terms present in the correct answer but absent from the model’s output. When triggered, we re-issue the original prompt augmented with the model’s prior answer and reasoning and an explicit CLINICAL KNOWLEDGE CORRECTION block listing the missed concepts, and ask the model to re-examine the image in light of this correction. We record whether the model changed its answer (*adaptation_changed*) and whether the revised answer is correct (*adaptation_correct*), and report the *adaptation rate* – the fraction of triggered questions that the model corrects to the right answer. This distinguishes two failure modes that look identical from accuracy alone: a genuine *knowledge gap*, where the model cannot incorporate the correction even when handed it explicitly, versus an *activation failure*, where the model possessed the relevant knowledge but did not surface it unprompted.

3.3 Soft Gating

Because the five phases form a causal chain, scoring a diagnosis (DSCR) as valid reasoning after a model has already failed to identify the correct anatomy (AIA) produces a misleading result – the diagnosis was not built on correct upstream perception. Soft gating enforces this directly: a phase is blocked (*gated_out*, scored 0.0) unless every phase it depends on scored at or above a gating threshold τ_c (default 0.5). AIA has no upstream dependency and always runs; LIL requires AIA; DSCR requires AIA and LIL; PJRF requires AIA, LIL, and DSCR; TCM requires all four. *Chain completion rate*, reported at the case level, is the fraction of cases in which every phase ran and cleared its gate – a model with a high chain completion rate is demonstrating coherent end-to-end reasoning rather than isolated per-phase competence. Gating can be disabled for a flat accuracy baseline.

4 Chain Validity in OmniBrainBench

The KAB framework as described in Section ?? presupposes that a case’s five phases represent one patient’s actual clinical trajectory: a model that fails AIA for a given patient should be gated out of that same patient’s DSCR question, not an unrelated patient’s. We investigated whether OmniBrainBench (?) supports this premise directly, by checking whether any image is shared across questions tagged with different clinical phases within a

grouped case – the only way genuine same-patient, multi-phase continuity could hold.

Of 6,823 total questions drawn from 15 source sub-corpora (including PubMedVision, BraTS2021, RSNA, ISLES2022, and VQA_RAD, among others), we found only 2 questions that genuinely share a same-patient image across two different phases, both from VQA_RAD and both AIA-LIL pairs. We repeated the analysis with a second, independent method – extracting real per-patient case identifiers by stripping known modality/view/slice suffixes from filenames on a per-corpus basis – and obtained the same result: 2 genuine multi-phase same-patient cases in the entire dataset, zero spanning three or more phases, zero spanning all five. The root cause is structural: each of the 15 source sub-corpora was constructed for one specific clinical task (e.g., BraTS for tumor characterization, RSNA for hemorrhage classification, ISLES for stroke), so a given real patient within a sub-corpus is essentially only ever asked one phase-type of question. The grouping key currently used to construct multi-phase "cases" (*source_file*) in practice groups together an entire origin sub-corpus, standing in for a single patient; a "chain" evaluated under this grouping is stitched from *different real patients*’ questions, not one patient’s actual diagnostic journey. A model gated out of a later phase after failing an earlier one under this grouping demonstrates nothing about causal reasoning degradation for any single patient.

We searched for a replacement or supplementary dataset with genuine same-patient, multi-phase continuity and a ready-made question layer (Section ??); none exists. This motivates the construction described next.

5 Constructing a Genuine Causal-Chain Dataset from LUMIERE

We build a genuine same-patient longitudinal chain dataset on LUMIERE (?), a single-center dataset of 91 real glioblastoma patients with 638 study dates, automated multi-sequence tumor segmentation, and clinician-assigned RANO treatment-response ratings with documented rationale. Figure ?? summarizes the automatic construction pipeline described in this section; the domain-expert review pass that finalizes each item runs as a separate, ongoing track (Section ??) and is not part of this pipeline diagram.

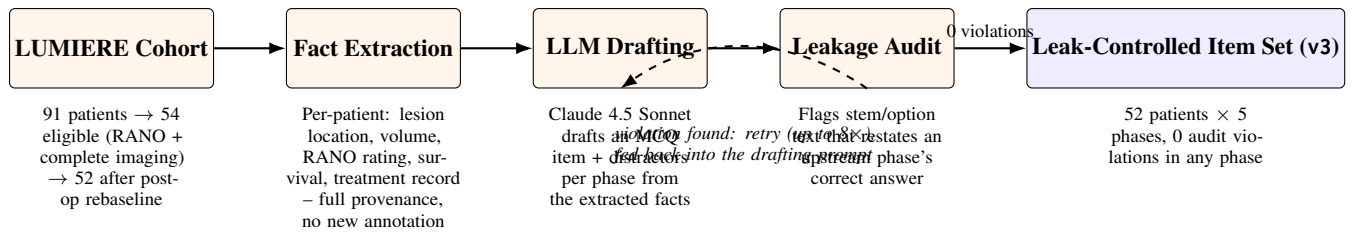


Figure 2: Construction pipeline for the leak-controlled (v3) LUMIERE item set. Structured per-patient facts are extracted from LUMIERE, an LLM drafts multiple-choice items from those facts, and an automatic auditor flags any item that restates an upstream phase’s correct answer, feeding violations back into the drafting prompt for up to eight retries. The domain-expert review pass that finalizes each item (Section ??) is a separate, still-ongoing track and is not shown here.

Fact extraction. For each patient, we extract per-phase facts directly from LUMIERE’s shipped data with full provenance, requiring no additional annotation for the perception-oriented phases: LUMIERE’s tumor segmentation masks are already registered to a standard MNI152 atlas space (the same space used by standard neuroanatomical atlases), so lesion-location facts reduce to a voxel-index lookup rather than a registration step, and per-timepoint tumor volumes ship precomputed. DSCR facts use LUMIERE’s existing expert-assigned RANO ratings directly rather than requiring new diagnostic annotation. PJRF and TCM facts draw on the dataset’s structured survival and Stupp-protocol treatment records.

Cohort. Of LUMIERE’s 91 patients, 55 have at least one timepoint with both a documented RANO rating and complete multi-sequence imaging, the minimum requirement for a phase-complete case; we draft from this full 55-patient eligible cohort rather than a smaller convenience sample so that a domain-expert review pass can cover the entire usable population in one sitting. One patient (Patient-025) has no timepoint satisfying both conditions simultaneously and is dropped at extraction time, leaving **54 patients** and 269 drafted multiple-choice items in the item set described below.

Timepoint sampling. We initially selected each patient’s most recent RANO-rated timepoint, but found this made nearly every DSCR answer "progressive disease," since glioblastoma patients overwhelmingly trend toward progression by their final scan – a sampling choice that would have made the DSCR phase trivially guessable regardless of a model’s actual reasoning. We instead sample a random eligible response-rated timepoint per patient (fixed seed), which yields a substantially more balanced label distribution across the 54-patient co-

hort: 37 progressive disease, 9 stable disease, 4 partial response, and 4 complete response. We additionally found and fixed a related bug in which a randomly sampled RANO-rated timepoint could lack matching complete imaging, silently falling back to imaging from a *different* date than the one the RANO rating and DSCR facts describe; we now constrain sampling to timepoints with both a rating and complete imaging, and verified zero such mismatches across the full extracted cohort.

Hybrid drafting and expert review. A large language model (Claude 4.5 Sonnet, accessed via a local LLM proxy) drafts multiple-choice items per phase from the extracted per-patient facts, including plausible distractors. Every drafted item is then reviewed by a domain-expert clinical collaborator before it is admitted to the dataset: the reviewer can approve an item as-is, edit the question text or any answer option (including which option is marked correct) before approving, or reject an item outright with a required justification. Items the drafting process itself flags as an ambiguous case or a low-confidence distractor are surfaced to the reviewer with a visual marker for closer attention. This two-stage construction is deliberate: the perception-adjacent phases (AIA, LIL) and the diagnosis phase (DSCR) are close to programmatically derivable from LUMIERE’s shipped segmentation and RANO data, but constructing plausible-but-wrong distractors for the prognosis (PJRF) and treatment (TCM) phases requires judgment informed by the glioblastoma prognostic and treatment literature – particularly around confounds such as pseudoprogression, which is a well-known source of misjudgment even for practicing radiologists that we do not consider safe to leave to an LLM alone.

Review mechanism. Because the domain-expert reviewer is a clinical collaborator without cluster or command-line access, the review interface is a single self-contained HTML file bundling every drafted item together with its supporting MRI slice images, requiring no server, account, or installed software – the reviewer opens it directly in a browser, and their decisions are saved locally and returned as an updated copy of the same file.

Preliminary evaluation on unreviewed items. The expert review pass described above validates every item before it is treated as a final benchmark, but it is not, on its own, a precondition for using the drafted items to test the KAB framework and probe the models’ behavior: with the senior author’s approval, we run preliminary evaluations directly on the LLM-drafted items while the review pass is in progress, and treat the two as independent tracks rather than a strict pipeline. This choice is what surfaced the item-quality problem described next – a problem the review pass alone, applied only after drafting, would not have caught before a substantial reviewer time investment.

Stem-leakage audit and a leak-controlled item set. Manual inspection of the drafted items surfaced a systematic construction flaw: the drafter wrote each phase’s question stem as a self-contained clinical vignette, which led it to restate findings established in earlier phases of the same chain. We built an automatic auditor that flags, per phase, any stem or option text containing content it should not have access to – a numeric percent volume change, a raw volume or region name, a change-direction word, a RANO label, or a stated survival duration – checked both against the phase’s own correct answer (*self-leakage*) and against the correct answers of upstream phases in the same chain (*upstream leakage*). Applying this auditor to the initial item set (v2) found leakage in a majority of DSCR, PJRF, and TCM items and a smaller fraction of LIL items; a lexical baseline that guesses the option with the highest word overlap with the stem performs at or below chance on the same items, confirming the leak is semantic (e.g., a stated percent volume increase implying "progressive disease") rather than a copy-paste artifact a naive overlap check would already catch.

We traced the cause to the drafting prompt itself, which supplied each phase’s drafting call with the correct answers of upstream phases – necessary for constructing a coherent chain – without

instructing the model not to restate them, and, for DSCR specifically, explicitly asked the model to let the reader "infer" the response label from volume-change facts it had just been shown. We built a revised, leak-controlled item set (v3) that keeps the same upstream-truth context available to the drafter but forbids restating it in the visible stem or options, re-drafting automatically (up to eight attempts) whenever the auditor flags a violation and feeding the specific violated rule back into the retry prompt. The DSCR stem, which the auditor showed was hardest to keep leak-free under free-form drafting, is instead built from a fixed template populated only from structured extent-of-resection, re-resection, and radiotherapy-timing fields, with no correct-answer information available to leak. The leak-controlled item set clears the auditor with zero violations across every phase.

Post-operative rebaselining. Building the leak-controlled item set surfaced a second issue upstream of leakage: the LIL phase’s volume-change baseline was, for 42 of 54 patients, a pre-operative scan, so the measured "volume change" mostly captured the surgical resection rather than the treatment response that the RANO rating (DSCR) is actually judged against. We recompute the LIL baseline as each patient’s latest imaged post-operative scan prior to the follow-up (advancing past any re-resection that occurs before the follow-up date), changing the baseline for 44 of 54 patients and shifting the cohort’s volume-change direction from a resection-dominated skew to a more balanced 20 patients increasing versus 32 decreasing. Two patients (Patient-049, Patient-068) lack a post-operative scan with matching segmentation and are excluded from the leak-controlled item set rather than mixed in on a pre-operative baseline, leaving 52 patients. Downstream DSCR, PJRF, and TCM facts and answer keys, which do not depend on the LIL baseline, are unchanged by this step.

Display-orientation note. Rendered slices follow radiological convention (patient’s right shown on the image’s left), which the original prompt never stated explicitly. For the leak-controlled item set only, we prepend a one-line statement of this convention to the image content sent to the model, leaving OmniBrainBench and the initial LUMIERE item set unaffected.

Contribution. Once constructed, this is – to our knowledge – the first dataset combining genuine

same-patient continuity with a full imaging-to-treatment question layer, closing a gap that persists across every existing candidate we are aware of (Section ??). Beyond the dataset itself, we position the hybrid drafting-plus-targeted-expert-review pipeline as a reusable construction method: it is a substantially cheaper way to turn a raw longitudinal clinical dataset into a validated multi-phase reasoning benchmark than a fully manual annotation effort, while still routing the judgment calls that matter clinically (prognosis and treatment distractors) through expert review rather than LLM drafting alone.

6 Experimental Setup

Models. Our model suite was revised after the framework was first designed: the senior author restricted the roster to models with no China-developed origin, which removed five open-weight models from an earlier 12-model plan (Qwen2.5-VL, Qwen3-VL-30B, InternVL3-38B, HuatuoGPT-V-34B, and Lingshu-32B) and, independently, we dropped Llava-Med-7B after confirming it cannot reliably produce the structured JSON output KAB’s parsing requires even under a guided-decoding retest. The current suite spans two categories: three proprietary models accessed via API or a LiteLLM proxy (GPT-5, Claude 4.5 Sonnet, Gemini 2.5 Pro), and four open-weight models served through Ollama at 4-bit quantization (MedGemma-4B, the suite’s only medical-domain fine-tune; and three general-purpose models, Gemma-3-12B, Gemma-3-27B, and Llama-4-Scout, a 109B-parameter mixture-of-experts model with 17B active parameters). All models are accessed through the same OpenAI-compatible interface so the evaluation pipeline is identical regardless of backend.

Infrastructure. Open-weight model serving and inference runs on single-GPU SLURM jobs (NVIDIA RTX PRO 6000). The roster revision above has an infrastructure consequence worth stating explicitly: the removed models included every open-weight model in the 30B-38B range that would have required multi-GPU tensor parallelism, so every model in the current open-weight suite serves at 4-bit quantization on a single GPU, and the tensor-parallelism question we anticipated in earlier planning does not arise for this suite.

Knowledge bases. RadLex (RSNA, English-filtered OWL release) and the NCI Thesaurus (NIH NCI, OWL release) are used as-is, with no stub or fallback vocabulary: the KB aligner requires both files to be present and raises an error rather than silently degrading if either is missing.

7 Testing Whether Correct Answers Are Image-Grounded

Faithfulness (Section ??) tests whether a model’s stated reasoning is internally consistent with its answer; it does not test whether that reasoning is actually derived from the image. The local faithfulness probe in particular is a weak instrument for this specific question by construction: it asks the same model to predict its own answer from its own reasoning text, so any response whose reasoning happens to restate the chosen option passes the probe regardless of whether an image was ever consulted, independent of how often that occurs in practice. We therefore built a separate suite of four checks, run on both the initial and leak-controlled LUMIERE item sets (Section ??) and applied identically across the full model suite (Section ??), to test image-grounding directly rather than inferring it from a self-consistency probe.

Stem-leakage audit. Described in Section ?? as the mechanism used to build the leak-controlled item set, the same auditor doubles as a grounding diagnostic: an item whose stem or upstream chain context already states the correct answer in words is not testing image use at all, regardless of what a model’s accuracy or faithfulness score reports. Running the two item sets through this single, unmodified auditor therefore also serves as a controlled before/after manipulation for the three checks below.

Text-only ablation. We re-run the evaluation pipeline with all images stripped before the main call (question stem, answer options, and chain context from upstream phases are otherwise unchanged), and compare accuracy against the matched image-present run on the same items. Both arms run with gating disabled, so every phase is scored for every patient regardless of upstream correctness; gating a phase’s score would not change what a model is shown or answers, only whether that answer counts, and disabling it here avoids shrinking the comparison sample on later phases. The comparison isolates a specific claim:

if text-only accuracy is not distinguishable from image-present accuracy, the image is not necessary for the model to reach its answer on this item set, whatever its stated reasoning claims.

Evidence fact-check. Necessity and use are different claims – an image could be unnecessary on average while still being consulted – so we separately audit the content of a model’s stated evidence against the same structured per-patient facts used to draft the items (Section ??). For each factual claim in a correct answer’s `visual_grounding` and `reasoning` fields (hemisphere/region statements and any measurement with units), we assign one of four labels: *copied*, the claim’s content already appears in the stem, answer options, or the upstream chain context the model was shown; *independently supported*, the claim was not shown to the model but matches the ground-truth facts; *fabricated*, the claim matches no ground-truth fact; or *laterality flip*, a hemisphere claim that disagrees with the segmentation under the assumption that image left corresponds to patient right. We score the fourth category separately from ordinary fabrication because the rendered slices follow radiological display convention without stating so in the prompt for the initial item set (fixed for the leak-controlled set, Section ??): a hemisphere claim that is wrong under one convention and correct under the other reflects a display-convention ambiguity in the stimulus, not a perceptual error, and conflating the two would misattribute a prompt-design gap to the model. RANO-criteria thresholds quoted verbatim from clinical guidelines (e.g., “ $\geq 25\%$ increase”) are excluded from the fabricated-measurement count, since they restate a rule rather than assert a patient-specific fact.

Counterfactual-image substitution. The strongest of the four checks: rather than removing the image (ablation) or auditing what a model says about it (fact-check), we substitute a *different, real patient’s* own image in place of the original patient’s, while leaving every text field – stem, answer options, chain context, and the image labels naming which timepoint is baseline versus follow-up – exactly as the original patient’s. A model that is actually using the image should be more likely to answer consistently with what is now in front of it; a model relying on the (unchanged) text should not. We pair every patient in the leak-controlled item set with a partner via a deterministic, seeded assignment: because the item set’s post-operative

rebaselining (Section ??) yields a near-even split of patients whose lesion volume increases versus decreases between baseline and follow-up, every patient can be assigned a partner with the *opposite* volume-change direction, guaranteeing a substantively different image is shown in every case; we additionally report, without optimizing for it, how often the pairing also yields a different DSCR response-assessment label. The substitution is applied to all five phases for completeness, using each partner’s own rendered timepoint images for the phase in question, and evaluated non-gated for the same sample-preservation reason given above. Our primary comparison is the LIL and DSCR phases, the two phases whose correct answer is most directly a function of image content; AIA’s correct answer does not vary with the image shown (a single fixed sequence-protocol string across the cohort) and is included only for completeness, and PJRF/TCM’s own-image effect is confounded by upstream phases that are simultaneously evaluated under their own swapped image, so a chain-context effect from an upstream swap cannot be fully separated from a same-phase effect there.

Our primary outcome is the *flip rate*: the fraction of items on which a model’s answer under the substituted image differs from its answer under the patient’s own image. We additionally check, among items that do flip, whether the new answer matches what the substituted patient’s own facts would actually support, which distinguishes a flip driven by genuine image content from one driven by decoding variance alone. We pre-registered the joint reading of all four checks before running any of them: text-only accuracy statistically indistinguishable from image-present accuracy, evidence classified mostly as copied rather than independently supported, and a low, non-truth-tracking counterfactual flip rate together would indicate that correct answers on this item set are not meaningfully image-grounded; a clear text-only accuracy drop together with independently-supported evidence and a counterfactual flip rate that tracks the substituted patient’s truth would instead indicate genuine grounding, in which case we would not report a non-grounding claim regardless of what the first three checks alone suggested.

8 Results

We report results from three of the four grounding-test checks described in Section ??: the stem-

Model	Image	Text-only	Δ
Llama-4-Scout	49%	52%	+3
Gemma-3-27B	54%	50%	-3
Gemma-3-12B	48%	51%	+3
MedGemma-4B	33%	36%	+3

Table 1: Non-gated overall accuracy on the leak-controlled (v3) item set, with images present versus text-only. Δ = text-only – image, in percentage points.

leakage audit, the text-only ablation, and the evidence fact-check. All numbers below use the leak-controlled (v3) item set unless stated otherwise, so that a leftover leakage effect cannot be mistaken for a grounding effect. The fourth check, counterfactual-image substitution, was still running across the model suite at the time of writing and is not reported here (Section ??); these results should accordingly be read as provisional, not as the paper’s final grounding claim.

Stem-leakage audit. The initial item set (v2) failed the auditor at rates too high to treat its accuracy numbers as informative about image use: 0% of AIA items leaked, but 13% (7/54) of LIL items self-leaked, 98% of DSCR items restated an upstream phase’s exact percent volume change, 85% of PJRF items either stated the patient’s actual survival duration directly (31%) or named the upstream RANO label (72%), and 98% of TCM items restated LIL’s volume-change percentage or RANO label. The leak-controlled item set (v3), built with the same auditor folded into the drafting retry loop (Section ??), cleared every phase with zero violations. All results below use v3.

Text-only ablation. Table ?? compares non-gated overall accuracy on v3 with and without images, for all four open-weight models. Three of four models score at least as high without the image as with it (Llama-4-Scout, Gemma-3-12B, and MedGemma-4B each +3pp); Gemma-3-27B is the only model with an image advantage (–3pp text-only), driven mainly by AIA (86% text-only versus 77% image), and even that gap falls inside the 95% Wilson confidence interval at $n = 52$. No model shows an image advantage outside its own confidence interval on any phase. At the level of individual questions rather than aggregate accuracy, the pattern is sharper still: the same question receives an identical answer with and without the image for 87-91% of items, for every model. Among the minority of items where removing the image

changes the answer, "right only with image" and "right only without image" occur at comparable rates (55 versus 67, summed across models) – the pattern expected if the image swap behaves like decoding noise rather than the removal of information the model actually needed.

For context, v3 items are substantially harder than v2 for every model (image-present overall accuracy down 13-19 percentage points, e.g., Llama-4-Scout 67%→49%, MedGemma-4B 45%→33%), consistent with the stem-leakage fix removing an exploitable shortcut rather than removing task difficulty. This drop is a validity correction to the earlier item set, not itself evidence about image use; we report it only to establish that the narrow with/without-image gap on v3 is not an artifact of v3 being an easier, less discriminating test.

Evidence fact-check. Restricting to correct answers on v3, the fraction of stated visual_grounding/reasoning claims classified copied – already present in the stem, answer options, or chain context the model was shown – ranges from 43% to 90% across the four models. The complementary category that would demonstrate genuine image use, independently supported, accounts for only 0-8% of claims. Gemma-3-27B is an outlier within this pattern: rather than copying, it more often states no checkable claim at all, which is consistent with, not contrary to, the same underlying picture – its stated evidence rarely contains a fact it could only have obtained from the image either way.

Two findings that sharpen this reading. Two additional v3 results bear on how the numbers above should be read, in different directions. First, a trivial majority-class baseline – always answering "progressive disease" – scores 64-65% on DSCR, because LUMIERE’s glioblastoma cohort is overwhelmingly progressing and this is both the correct answer and typically the longest option; all four models sit at or below this baseline (Gemma-3-27B 65%, Llama-4-Scout 58%, Gemma-3-12B 56%, MedGemma-4B 14%). This is a construct-validity caveat on the DSCR numbers specifically – no model demonstrates RANO-assessment skill above what the class imbalance alone would predict – and should not be read as further evidence for or against image grounding. Second, LIL accuracy is at or near the four-option chance rate of 25% for three of four models (Gemma-3-12B 23%, Gemma-3-27B 27%, Llama-4-Scout 25%; MedGemma-4B

40%), and text-only LIL accuracy is statistically the same (23-38%). Unlike the DSCR finding, this result speaks directly to grounding: models are not merely indifferent to the image on LIL, they are unable to extract the lesion volume-change direction from it even when the task asks for exactly that comparison.

Reading against the pre-registration. Section ?? pre-registered that text-only accuracy statistically indistinguishable from image-present accuracy, together with evidence classified mostly as copied rather than independently supported, would indicate that correct answers on this item set are not meaningfully image-grounded. The three completed checks meet both conditions on v3: text-only accuracy is within the confidence interval of image-present accuracy for every model on every phase, and independently-supported evidence accounts for only 0-8% of correct answers' stated claims. We read this as satisfying the paper's own pre-registered criterion for "claim supported," with the two qualifications already stated in Section ??: these numbers come from the LLM-drafted item set ahead of its domain-expert review pass, and the fourth, strongest check – counterfactual-image substitution – was still running across the full model suite at the time of writing. We treat the reading above as provisional until both resolve, rather than as this paper's final grounding claim.

Limitations

This paper reports the KAB framework, the chain-validity problem it surfaces in an existing benchmark, the dataset construction method used to address it, the image-grounding test suite of Section ??, and provisional results from three of that suite's four checks (Section ??); it does not yet report full evaluation results across the model suite described in Section ?. Two items are in progress at the time of writing. First, the LUMIERE-based item set (Section ??) is fully constructed and drafted, and preliminary evaluations run directly on the unreviewed, LLM-drafted items with the senior author's approval, but the domain-expert review pass that would validate every item as final is still ongoing; any preliminary result drawn from the unreviewed items, including those in Section ??, should therefore be read as provisional, and is exactly the class of item-quality problem that motivated the grounding-test suite of Section ?? in the first place. Second, of the four checks in that

suite, the counterfactual-image substitution – the strongest of the four, and the only one not yet reported – had not finished running across the full model suite at the time of writing. We view withholding a causal-degradation or grounding claim until both are resolved as the responsible ordering of claims, rather than treating the provisional reading in Section ?? as final ahead of the check designed to corroborate or narrow it.

Several additional limitations apply to the framework and dataset construction as currently specified. First, KB alignment (Section ??) is a string- and n-gram-matching metric against a fixed vocabulary, not a semantic one; it will undercount a model that expresses a correct clinical concept in phrasing not present in either ontology, and we expect this to be a point of reviewer scrutiny. Second, PJRF and TCM are a small fraction of any brain-MRI MCQ dataset assembled from real clinical sources (well under 5% of cases), so results on these two phases carry low statistical power and should be reported with confidence intervals rather than pooled with the larger phases, which would mask exactly the degradation signal the framework is designed to detect. Third, the tumor segmentations underlying LUMIERE's lesion-location and volume facts are produced by automated tools (DeepBraTumIA, HD-GLIO-AUTO) and are not manually verified per-timepoint, whereas the RANO treatment-response ratings are expert-assigned; we state this distinction explicitly rather than implying uniform annotation quality across fact types. Fourth, LUMIERE is a single-center cohort, and a benchmark built on it inherits whatever institution-specific imaging protocol and patient population characteristics that entails. Fifth, the local faithfulness probe (Section ??) is, by construction, only able to detect an inconsistency between a model's stated reasoning and its answer; it cannot on its own establish that consistent reasoning is image-derived, which is why Section ?? tests grounding through ablation, evidence auditing, and counterfactual substitution rather than through faithfulness alone. Finally, the model suite in its current, roster-revised form (Section ??) no longer includes any open-weight model requiring multi-GPU tensor parallelism, so the hardware constraint anticipated in earlier planning does not apply to the results reported here; a future roster reintroducing a larger open-weight model would reopen that question.

Ethics Statement

This work evaluates the diagnostic reasoning of general-purpose and medical-domain language models on real, de-identified patient imaging data; it does not deploy, recommend, or endorse any model for clinical use, and no results in this paper should be interpreted as a clinical validation of any evaluated system. LUMIERE is used under its published non-commercial research license, and all patient data is accessed in its already de-identified, publicly released form; no new identifiable patient information is collected. The domain-expert review step (Section ??) is conducted by a qualified clinical collaborator under the direction of the senior author, and reviewer decisions are used only to validate or correct machine-drafted question content, not as a source of new clinical diagnoses.

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